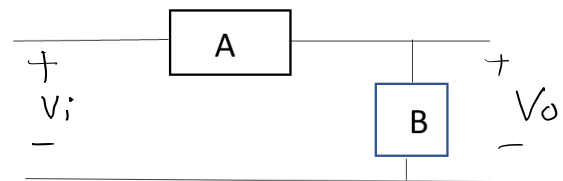
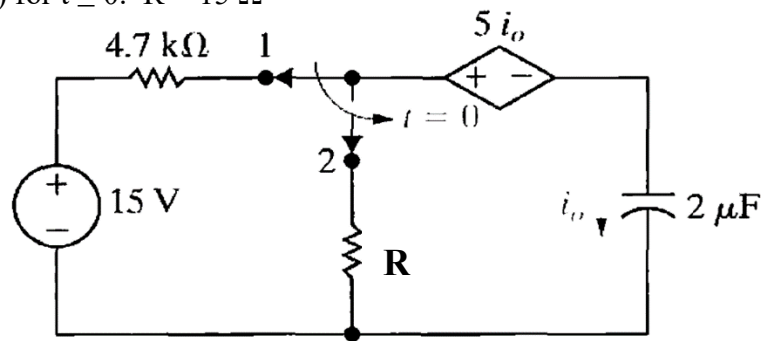


1. A system described by the equation $\dot{x}_1 = -x_1 - x_2$ and $\dot{x}_2 = x_1$ has damping factor ζ and natural frequency ω_n respectively
 (a) 0.5 and 1 (b) 1 and 1 (c) 0.707 and 2 (d) 1 and 2
2. Number of equilibrium points of the system described by the following equations
 (a) 1 (b) 2 (c) 3 (d) 4
3. The equilibrium point (0,0) of the population system described by the equation $\dot{x}_1 = x_1 - x_1 x_2$ is
 $\dot{x}_2 = -x_2 + x_1 x_2$
 (a) a stable focus (b) saddle point (c) an unstable node (d) an unstable focus
4. In the above system for nonzero population the state trajectory will circle around the point
 (a) (1,-1) (b) (0,0) (c) (1,1) (d) (-1,1)
5. In the circuit shown in figure shown A consists of resistance of 1Ω and an inductance of $2H$ in parallel. B consists of a resistance of 1Ω and a capacitance of $0.5 F$ in parallel. Voltages V_i and V_o are the inputs and outputs respectively. Find the time constant of the circuit.



6. The switch in the circuit has been in position 1 for a long time before moving to position 2 at $t = 0$. Find the current $i_o(t)$ for $t \geq 0$. $R = 15 \Omega$



7. What is the output of the MATLAB code in Fig. a. The function used is shown in Fig. b [6]

```
ts=[0 4]; x0=[0;0];
[t,x]=ode23('quiz',ts,x0);
[a,b]=size(x);
x(a,:)
```

Quiz.m

```
function c= quiz(t,x);
c=[x(2);-2*x(1)-3*x(2)+1];
```

8. Draw the approximate trajectories around the equilibrium points for the circuit described by the differential equation $\ddot{x} + 0.5\dot{x} + 2x + 0.2x^2 = 0$
9. Write the answers in one sentence
1. What is the difference between limit cycles and oscillations in undamped pendulum?
 2. What is meant by 'chaos' in system studies?
 3. How can we obtain different nonlinear phenomena in Chua's circuit experimentally?
 4. Why is the extinction of species in impossible in prey-predator mode?